Ganesh Chandrasekar

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https://github.com/cbsag portfolio-cbsag-2026.vercel.app

PROFILE

Master's thesis student in Computer Science at Concordia University, working on biomedical NLP, retrieval-augmented generation, and evidence-grounded question answering. My experience brings together applied AI research and software development, from building RAG pipelines in the lab to working on full-stack and cloud-based product systems. I enjoy building practical AI tools that turn research ideas into systems people can test, use, and improve.

EDUCATION

Master of Computer Science (Thesis) — 09/2023 – present
Concordia University — Montreal, Canada

Bachelor of Engineering — 2018 – 2022
Anna University (KCG College of Technology) — Chennai, India
B.E. Computer Science and Engineering - **CGPA 8.81/10** (**First Class with Distinction**)

PROFESSIONAL EXPERIENCE

Research Assistant — 09/2023 – present
CLaC Lab, Concordia University — Montreal, Canada

- Built applied AI research prototypes for **biomedical question answering**, **information retrieval**, and **retrieval-augmented generation**.
- Designed modular **RAG pipelines** combining **sparse** and **dense retrieval**, **reranking**, **evidence selection**, **prompt engineering**, and **grounded LLM generation**.
- Developed systems for **MedHopQA** and **TREC BioGen Task B**, focusing on **evidence-grounded** biomedical answers with **traceable citations**.
- Optimized **LLM inference** and experimentation workflows on **HPC** infrastructure using **Slurm**, **Conda**, and **vLLM**.
- Collaborated with researchers to turn research ideas into working pipelines, evaluation outputs, and analysis tools.

Teaching Assistant — 09/2024 – 05/2026
Concordia University — Montreal, Canada

- Teaching Assistant for COMP 479/6791 – **Information Retrieval and Web Search** and COMP 232 – **Discrete Mathematics**.
- Support students with coursework, assignments, project guidance, and core concept clarification.
- Handle grading and course responsibilities accurately while meeting deadlines.

OrangeScape Technologies Pvt Ltd — 07/2022 – 08/2023
Solution Developer — Chennai, India

- Worked as a Solution Developer on **Kissflow Procurement Cloud**, contributing to **frontend**, **backend**, **workflow automation**, and product feature development.
- Built and integrated **backend scripts**, **APIs**, and workflow components within a **Low-Code/No-Code** product environment.
- Worked with cloud-based development workflows, **Docker-based environments**, and **SQL/NoSQL** data systems.
- Collaborated with product and engineering teams to understand requirements, debug issues, test features, and deliver reliable software updates.

Basik Marketing Pvt. Ltd — 05/2022 – 07/2022
Software Engineering Intern (Backend) — Chennai, India

- Collaborated with the platform development team for **The Esports Club**, a marketing firm supporting the Indian gaming and esports industry.
- Developed backend components and data-handling logic to improve platform functionality and reliability.
- Evaluated platform requirements and supported implementation, testing, and debugging of backend features.

Solarillion Foundation 01/2021 – 01/2022
Undergraduate Research Assistant

- Developed a Two-Stage Machine Learning Engine for **Flight Delay Prediction**.
- Worked on **Citation Intent Classification** and supervised text analysis tasks.
- Supported data preprocessing, model experimentation, and research-oriented implementation.

PROJECTS

TREC BioGen: Biomedical Evidence-Based Generation Systems | Dashboard 🔗 2025

- Built an end-to-end **biomedical RAG** prototype combining **BM25, TF-IDF/MMR, MedCPT, cross-encoder reranking, and Qwen-based LLM** generation.
- Designed **evidence selection** and **citation-control** mechanisms so that generated answers only cite retrieved **PubMed** evidence.
- Optimized retrieval and generation stages to reduce context size while maintaining grounded biomedical answer quality.
- Created an interactive dashboard to analyze retrieval behavior, citation usage, and system outputs across questions.

Agentic and Non-Agentic Medical Question Answering Systems 2025

- Developed **agentic** and **non-agentic multi-hop QA systems** using **open-source LLMs**, **retrieval tools**, and structured **prompting**.
- Integrated **Wikipedia** and **PubMed retrieval** with **LLM-based reasoning**, **answer generation**, and **short-answer extraction**.
- Compared **tool-calling agents** with controlled pipeline-based systems to study reliability, evidence use, and answer quality.

PROGRAMMING LANGUAGES AND SKILLS

Programming Languages

- Python, JavaScript, Java, SQL

AI / ML / NLP

- PyTorch, Transformers, vLLM, scikit-learn, Pandas, Retrieval-Augmented Generation, LLM pipelines.
- Prompt Engineering, Reranking, Evidence Selection, BM25, Pyserini/Lucene, FAISS, OpenSearch, MedCPT.

Full-Stack Development

- React JS, Node JS, Nest JS, Flask, REST APIs, Progressive Web Apps.

Cloud, Containers, and Infrastructure

- AWS, GCP, Docker, Linux, Conda, Slurm HPC.

Tools and Data Systems

- SQL, NoSQL, Git, GitHub, VS Code, Jupyter Notebooks, LaTeX, Figma, GATE, Codex, Claude Code.

PUBLICATIONS

Modular Sparse and Dense Retrieval for Evidence-Constrained Biomedical Question Answering 🔗 2025
G. Chandrasekar, B. L. Follo, A. Vinokhodov, S. Bergler
Proceedings of the Thirty-Fourth Text REtrieval Conference (TREC 2025), BioGen Track, National Institute of Standards and Technology (NIST), 2025. System paper describing modular sparse and dense retrieval pipelines for PubMed-grounded biomedical question answering with evidence-constrained LLM generation and PMID citation control.

Agentic and Non-Agentic Multi-Hop Systems for Medical Question Answering 🔗 2025
G. Chandrasekar, H. G. Saisudha, and S. Bergler
Proceedings of the BioCreative IX Challenge and Workshop (BC9): Large Language Models for Clinical and Biomedical NLP at the International Joint Conference on Artificial Intelligence (IJCAI), Montreal, Canada, 2025. Published Aug. 14, 2025. doi: 10.5281/zenodo.16875905

Exploration of Performance of Dynamic Branch Predictors used in Mitigating Cost of Branching 🔗 2022
G. Chandrasekar , A. Ambashankar, C. A. R, and S. S
Proceedings of the 2022 Third International Conference on Intelligent Computing Instrumentation and Control Technologies (ICICICT), 2022, pp. 574–579, doi: 10.1109/ICICICT54557.2022.9917915